

Study on Silent Guidance: Free Exploration as a Vector for Implicit Motor Adaptation

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Abstract

Learning new motor skills traditionally relies on directive instructions, which can limit spontaneous exploration and user autonomy. This study introduces the concept of "Silent Guidance," a paradigm where neuromotor adaptation is induced exclusively through the implicit manipulation of audio-visual affordances within an interactive digital environment. Fifteen healthy subjects, aged 21 to 63, interacted with a game ("Forest of Senses") controlled by a high-fidelity isometric sensor (Dextrain Manipulandum), without any explicit instructions regarding the goal to achieve. The results demonstrate major and significant motor adaptation along two axes: the emergence of digital individuation (net isolation force increasing from 2.88 N to 4.53 N, $p = 0.0020$), stimulated by a virtual fatigue constraint forcing motor rotation; and the convergence of force towards an invisible sub-maximal target (accuracy increasing from 43.4% to 60.4%, $p = 0.0054$). By proving that total agency can co-exist with strict biomechanical requirements, this study opens new perspectives for interactive design in post-stroke neurorehabilitation.

Keywords: Human-Computer Interaction, Neurorehabilitation, Biomechanics, Implicit Game Design.

1 Introduction

Learning new motor skills via digital interfaces traditionally relies on explicit tutorials or directive instructions. While this approach guarantees rapid cognitive understanding of a task, it paradoxically limits spontaneous sensorimotor exploration. Literature on engagement psychology [7] demonstrates that intrinsic motivation in virtual environments relies heavily on satisfying the need for autonomy. However, in the design of therapeutic interactive environments, the tendency remains to dictate movement rather than allowing the user to discover it.

Although implicit exploration is a common onboarding mechanic in playful game design [5] (e.g., open worlds guided by environmental architecture), we propose in this study to redefine and transfer this paradigm as a genuine clinical biomechanical tool: "Silent Guidance." We designed an interactive screen-based environment, controlled by a high-fidelity isometric sensor, acting as a matrix of implicit neuromotor constraints. The objective is to demonstrate that it is possible to finely redirect a user's motor effort by relying exclusively on free exploration and the manipulation of audio-visual affordances.

To guarantee total agency, the interaction protocol relies on a "Zero-Instruction" rule. Users are introduced to the system with a single open directive: *"With this sensor, you will control what happens in the world. Explore freely. There are 4 phases, the first is just a one-minute calibration phase."* This article aims to prove that this lack of instruction, coupled with a reactive biofeedback loop, forces the central nervous system to adapt along two major behavioral axes:

1. **The emergence of individuation:** The spontaneous inhibition of muscular co-activations to isolate single-finger control, indirectly stimulated by the obligation to perform motor rotation to bypass virtual fatigue.
2. **Sub-maximal convergence:** The abandonment of ballistic force in favor of targeted isometric regulation.

By validating the feasibility of this approach on a cohort of 15 healthy subjects, this study questions the capacity of interactive environments to "sculpt" motor behavior, opening perspectives for future applications in neuroengineering and post-stroke rehabilitation.

2 Related Work

Our approach lies at the intersection of implicit interface design and motor control biomechanics.

2.1 Implicit Design and Cognitive Load

The work of Hatzl et al. [4] on user onboarding demonstrated that low-complexity environments benefit from implicit cues rather than textual instructions, thus preserving agency. Perceiving the action opportunities of an interface (affordances) requires active exploration [2]. In our system, this exploration allows the user to deduce the physical laws through trial and error. The choice of a minimalist design (without a numerical UI) also relies on the recommendations of Nast et al. [6], who emphasize that simple visual feedback limits cognitive overload and promotes body awareness.

2.2 Embodied Interaction and Feedback Manipulation

When body movements synchronously modify a digital environment (Embodied Interaction), it becomes possible to manipulate sensory feedback to guide learning [3]. Adamovich et al. [1] emphasize that introducing distortions or gaps between actual movement and virtual feedback is a powerful lever for forcing the activation of targeted brain networks. Our system exploits this principle by introducing audio and visual feedback modulations and extinctions in response to maladaptive biomechanical actions.

2.3 Digital Individuation: A Biomechanical Justification

Human movement is naturally limited by neural and biomechanical couplings that force fingers to move together (synkinesis), a phenomenon quantified in healthy subjects by a physiological deficit in independence [8]. Xu et al. [10] demonstrated that the recovery of global grip strength and fine control (individuation) rely on distinct neural systems.

Térémetz et al. [9] confirm that isometric training specifically targeting this independence via a high-fidelity sensor outperforms conventional global therapies.

3 System Design and Methodology

3.1 Hardware and Software Interface

Interaction occurs via the Dextrain Manipulandum [9], an isometric sensor measuring the compression force of all five fingers at an algorithmic frequency of approximately 14 Hz. The interactive virtual environment, "Forest of Senses," features no text or visible numerical scores. The feeling of progression is conveyed by a poetic and implicit evolution of the ambient scenery. Progression through the four phases of the experience is dictated by a silent algorithm validating the accumulation of movements meeting the implicit constraints.



Figure 1: Photograph of the isometric hardware device (Dextrain) used for interaction.

3.2 The Four Phases of Constraint and Ambient Evolution

The virtual fatigue mechanic is active from Phase 1 and remains throughout Phase 2. It acts as a blocking energy gauge that forces the user into digital rotation to progress.

- **Phase 0 (Baseline):** Free exploration with no constraints to evaluate natural digital coupling. The visual environment reacts by displaying particle fountains whose behavior depends on the finger used and the pressure applied.
- **Phase 1 (Rotation and Volume):** The requirement for individuation is introduced in this phase via the Net Isolation Force calculation, coupled with a "winner-takes-all" rule (both remaining active through Phase 3): only the strongest finger drives the interaction, while being penalized by the activation of any parasitic fingers. Prolonged effort on this dominant finger increments the virtual fatigue gauge, gradually decreasing the volume of the audio-visual feedback, thereby forcing the

user to release the exhausted effector. Visually, a campfire lights up and the scenery evolves into a starry night.

- **Phase 2 (Isolation and Amplitude):** The system maintains the Net Isolation Force requirement and the implicit fatigue constraint from Phase 1. Furthermore, force amplitude now modifies the nature of the feedback to encourage intensity exploration: a weak press (0.5N to 4N) generates a single note and color; a medium press (4N to 8N) triggers two notes/colors; a strong press ($> 8N$) produces a three-note/color chord. Parasitic global pressing creates dissonance. Visually, day breaks.
- **Phase 3 (Spatial Convergence):** The user must reach and maintain a specific, invisible target force value. System reaction is modulated by the user's accuracy relative to this target: visually by the clarity of a ring, and aurally by the opening of a low-pass filter and the rhythm of familiar music (which adjusts to its original tempo when the correct force is applied). The ultimate reward for prolonged maintenance is the melting of the decorative snow.



Figure 2: Interface of the interactive environment in Phase 3. The visual ring acts as an implicit affordance guiding force convergence.

3.3 Participants and Procedure

This cross-sectional observational study included 15 healthy subjects recruited by convenience sampling. The cohort has a mean age of 27.6 years (range 21 to 63 years). The sample shows a strong predominance of right-handed subjects (14 using their dominant right hand, 1 left-handed using their dominant left hand). The interaction session lasted an average of 10 minutes per subject, introduced solely by the "Zero-Instruction" rule.

3.4 Extraction of Kinematic Metrics

High-frequency recording allowed the extraction of metrics upon which the algorithm reacts in real time:

- **Net Isolation Force:** Calculated using the formula $F_{max} - F_{second_max}$ (the force of the primary finger minus the force of the strongest parasitic finger). A high value indicates excellent inhibition of synkinesis. To exclude periods of motor rest and sensor noise, only time frames where a finger's maximum force exceeded 0.5 N

were kept for individuation analysis. Furthermore, Phase 1 was subdivided for visual analysis into three tiers (tertiles), allowing observation of exploration dynamics (Early Phase 1) and final adaptation (Late Phase 1).

- **Accuracy:** This parameter constitutes the primary measure of motor refinement during Phase 3. Its evaluation is carried out in two calculation steps.

Initially, the system determines the normalized absolute error E , defined by the following equation:

$$E = \frac{|\text{Force}_{\text{applied}} - \text{Force}_{\text{target}}|}{5} \quad (1)$$

(The denominator of 5 Newtons acts as a normalization constant, corresponding to a reference isometric effort empirically set during device calibration).

Subsequently, to ensure the biofeedback respects the users' neurophysiological reality, the final accuracy score P (ranging from 0 to 100%) does not derive from a linear subtraction. The E value is first mathematically constrained between 0 and 1, then injected into a non-linear transfer function (an interpolation spline), illustrated in Figure 3.

This curve defines an evaluation profile designed for rehabilitation:

- **Tolerance Zone (Initial Plateau):** For minimal deviations (E tending towards 0), the P score is maintained at 100%. This design choice allows for the absorption of natural motor noise (continuous isometric micro-tremors) without penalizing the user, thus preserving their agency.
- **Organic Degradation (Sigmoidal Slope):** As the error increases, the score reduction occurs fluidly, providing progressive and non-binary sensory feedback.
- **Critical Failure Threshold:** The accuracy score drops to zero ($P = 0\%$) when the force deviation becomes biomechanically unsuited for the targeted task (a threshold visually reached around $E \approx 0.8$ in Figure 3).

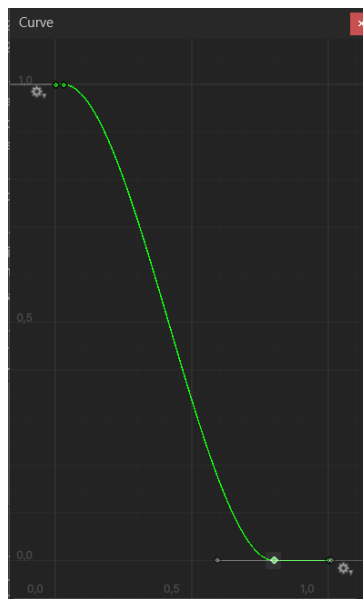


Figure 3: Transfer function curve defining the accuracy score (P) based on normalized error (E).

3.5 Ethical Considerations

This study was conducted in accordance with the fundamental principles of research involving human subjects.

- **Informed Consent:** Prior to the interaction, the objective of the experiment and the functioning of the Dextrain sensor were explained individually to all 15 participants. Oral consent was obtained from each subject, confirming their voluntary participation and their right to withdraw from the study at any time without justification.
- **Risk Assessment:** The required task was strictly non-invasive, consisting of a playful interaction with a virtual environment ("Forest of Senses") via an isometric force sensor. Given the nature of the exercise and the absence of sensitive medical data collection on a healthy cohort, the protocol presented minimal risk to participants.
- **Data Protection and Anonymization:** Although kinematic data were initially collected to allow for technical processing, a rigorous anonymization procedure was applied during the analysis phase. All identifying information was removed from the final result files, ensuring that the presented metrics (Net Isolation Force, Accuracy) could not be linked to any specific individual.
- **Institutional Framework:** In accordance with standard practices for non-clinical usability studies on healthy subjects within our institution, formal approval by an ethics committee was not deemed necessary for this proof-of-concept phase.

3.6 Qualitative Evaluation

Following the interaction, a questionnaire evaluated: the perceived goal of the experience, the identification of the rotation trigger, the awareness of isolation, and a Likert scale (1-5) assessment of the weight of visual and auditory modalities in motor adaptation.

3.7 Statistical Analyses

Intra-subject comparisons (Phase 0 vs. Phase 2, and Early P3 vs. Late P3) were performed using the non-parametric Wilcoxon signed-rank test, adapted to the size of our cohort ($N = 15$). The significance threshold was set at $p < 0.05$. The parametric magnitude of the adaptation was quantified using the effect size (*Cohen's d*) to facilitate clinical interpretation.

4 Results and Evidence of Adaptation

4.1 The Emergence of Individuation

The system induced significant inhibition of synkinesis. Using the Net Isolation Force metric, a clear increase is observed between free movement (Phase 0) and movement under implicit constraint (Phase 2). The overall cohort mean increased from 2.88 N (Phase 0) to 4.53 N (Phase 2), a progression of +1.65 N. Statistical analysis confirms a major and highly significant adaptation effect ($p = 0.0020$, $d = 0.99$). An effect size approaching 1.0

indicates that nearly the entire sample radically modified their biomechanical strategy in response to the feedback.

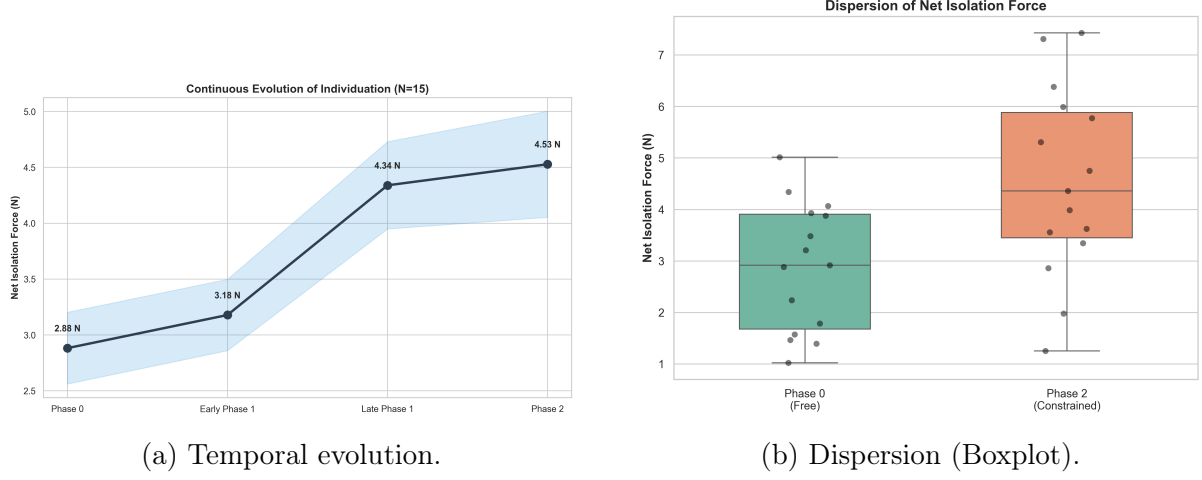


Figure 4: Improvement in Net Isolation Force between Phase 0 (Baseline) and Phase 2 (Isolation required).

4.2 Sub-Maximal Convergence (Phase 3)

The curve used to modulate the system's reaction in Phase 3 only rewards the user if they are within a specific force window around the target. Geometrically, this tolerated resonance zone represents a tiny percentage of the total effort amplitude permitted by the sensor. Consequently, a purely random pressing strategy would mathematically plateau around 15% accuracy.

However, the cohort achieved an overall mean accuracy of 54.6%, proving an intentional and directed search for the target. To overcome differences in execution time between subjects, time normalization was applied. Although digital rotation (induced by virtual fatigue) causes transient drops in accuracy to proprioceptively recalibrate pressure, a clear refinement of the overall motor strategy is observed: mean accuracy goes from 43.4% in the first quintile (initial 20%) of the phase, to 60.4% in the last quintile. This force smoothing is statistically significant ($p = 0.0054$, $d = 0.86$).

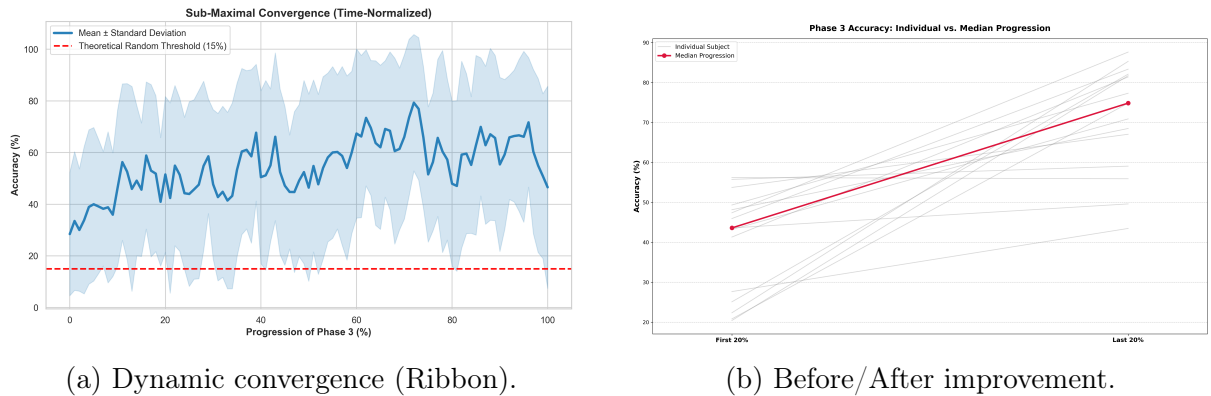


Figure 5: Evolution and tightening of accuracy in Phase 3, confirming the learning of a constant isometric force.

4.3 Qualitative Taxonomy of Exploration

Questionnaire analysis reveals that the vast majority of subjects (13 out of 15, or 86.7%) consciously perceived the isolation rule imposed by the algorithm. Furthermore, hearing emerged as the main proprioceptive vector for adaptation (the auditory indicator obtaining a mean influence score significantly higher than the visual indicator).

Cross-referencing kinematics and verbatim responses allowed the identification of three interaction profiles characterizing the diversity of exploration:

- **Profile A: Sensory Deductives.** They deduce the environment’s rules from the very first instances of penalization (*“Knowing how to use the right finger at the right time to do the right thing”*).
- **Profile B: Somatic Empiricists.** Highly exploratory, they test the sensor’s physical limits. Learning happens through confronting dissonance (*“I thought I could play chords at first but actually no,” “When there were several fingers at once it was a bit of a mess”*).
- **Profile C: Stochastic / Emergent.** Although they did not intellectualize the algorithm (e.g., *“No idea”* of the goal), recordings reveal an increase in their individuation. The environment therefore constrains biomechanics through action, even before clear cognitive formulation.

5 Discussion

5.1 Effectiveness of Silent Guidance

The study validates the hypothesis that "Silent Guidance" is an effective biomechanical constraint tool. By substituting explicit oral instructions with spatial and auditory manipulation of affordances, the system forced 15 naive subjects to inhibit their synkinesis and converge their effort, while preserving autonomy [7].

The protocol’s viability for the left-handed participant demonstrates the algorithm’s resilience, being agnostic to handedness. Moreover, the fact that "stochastic" profiles manage to develop excellent Net Force demonstrates the robustness of the approach: interaction shapes the right gesture, regardless of the cognitive strategy employed. Furthermore, the supremacy of the auditory modality in fine guidance suggests that future therapeutic environments would benefit from heavily relying on movement sonification to avoid ocular cognitive overload.

5.2 Limitations of the study

This study presents limitations inherent to its status as a proof of concept. The fixed sequential order of the phases does not allow isolating a potential habituation effect to the sensor. However, this order is consubstantial with our Game Design, acting as a progressive onboarding [4]. Secondly, although the system proved effective for the single left-handed participant, handedness could not be statistically compared in the overall analysis [8]. Finally, the short session evaluates short-term motor adaptation, which will be the subject of subsequent long-term clinical work.

6 Conclusion and Future Work

This study proves that total user agency is not incompatible with strict biomechanical requirements. It is possible to dictate a targeted neuromotor action to the human body through the simple arrangement of digital feedback and ambient rewards. The results obtained on healthy subjects clinically justify moving towards a future trial on post-stroke patients, where this approach could help overcome the motivational limitations of conventional neurorehabilitation.

Furthermore, a major evolution is envisioned within a 5-year perspective: the implementation of *Somatic Generative AI*. The hard-coded "Forest of Senses" environment could be replaced by real-time generative image and sound models. The system would no longer merely reduce volume upon error but would dynamically alter the harmony or visual style of the generated scene (e.g., shifting from a fluid watercolor style to a chaotic cubist style) based directly on the detected inhibition of synkinesis.

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